Goldman Sachs DSA Questions (Topic Wise Compilation)

Compiled from commonly reported interview experiences and preparation resources.

Arrays

• Two Sum
• 3Sum
• 4Sum
• Maximum Subarray
• Maximum Product Subarray
• Product of Array Except Self
• Best Time to Buy and Sell Stock
• Container With Most Water
• Trapping Rain Water
• Majority Element
• Missing Number
• Find Duplicate Number
• Merge Intervals
• Insert Interval
• Rotate Array
• Rotate Matrix
• Set Matrix Zeroes
• Spiral Matrix
• Kth Largest Element
• Subarray Sum Equals K
• Longest Consecutive Sequence
• Gas Station
• Jump Game
• Jump Game II

Strings

• Longest Substring Without Repeating Characters
• Longest Palindromic Substring
• Valid Anagram
• Group Anagrams
• Reverse Words in a String
• String Compression

- Decode String
- Valid Parentheses
- Generate Parentheses
- Minimum Window Substring
- First Unique Character
- Roman to Integer
- Palindrome Partitioning

Linked Lists

- Reverse Linked List
- Detect Cycle
- Merge Two Sorted Lists
- Merge K Sorted Lists
- Remove Nth Node From End
- Intersection of Linked Lists
- Copy List with Random Pointer
- Reorder List
- Palindrome Linked List
- LRU Cache

Trees & BST

- Level Order Traversal
- Diameter of Binary Tree
- Lowest Common Ancestor
- Validate BST
- Kth Smallest Element in BST
- Serialize and Deserialize Binary Tree
- Maximum Path Sum
- Symmetric Tree
- Path Sum

Graphs

- Number of Islands
- Clone Graph
- Course Schedule
- BFS
- DFS
- Topological Sort

- Dijkstra Algorithm
- Rotten Oranges
- Word Ladder
- Shortest Path in Binary Matrix

Dynamic Programming

- Climbing Stairs
- House Robber
- Coin Change
- Longest Increasing Subsequence
- Longest Common Subsequence
- Edit Distance
- Knapsack
- Partition Equal Subset Sum
- Decode Ways
- Unique Paths

Binary Search

- Search in Rotated Sorted Array
- Find Minimum in Rotated Sorted Array
- Peak Element
- Median of Two Sorted Arrays
- Koko Eating Bananas
- Capacity to Ship Packages

Heap

- Top K Frequent Elements
- K Closest Points to Origin
- Find Median from Data Stream
- Meeting Rooms II
- Task Scheduler

Backtracking

- Subsets
- Permutations
- Combination Sum
- N-Queens

- Sudoku Solver
- Word Search

Goldman Sachs Frequently Reported

- Search in Sorted Matrix
- Rotate Matrix by 90 Degrees
- Josephus Problem
- Valid Tic-Tac-Toe State
- Maximum Path Sum in Binary Tree
- Reverse Words in String
- Detect Loop in Linked List
- Minimum Flips for Alternating Binary String
- Dijkstra Shortest Path
- Trapping Rain Water
- Merge Intervals
- Number of Islands
- Coin Change
- LRU Cache